

Francesco Cavina

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Data and AI engineer with a BSc in Econometrics and Data Science from VU Amsterdam and incoming MSc studies at VU. Experience building privacy aware AI workflows, RAG systems, governed data interfaces, automated data pipelines, and full stack prototypes. Former VU teaching assistant with experience explaining abstract concepts clearly to first year students. Strong fit for educational technology work that combines practical AI development, clear communication, responsible GenAI use, and support for online and blended education.

TECHNICAL SKILLS

- **AI and GenAI:** LLM workflows, RAG, vector search, ChromaDB, OpenAI API, Hugging Face, sentence transformers, prompt design, citation based answers, local LLM workflows
- **Programming and data:** Python, SQL, PySpark, TypeScript, JavaScript, data pipelines, metadata driven ingestion, schema inference, data quality workflows
- **Web and product:** FastAPI, Next.js, React, REST APIs, Tailwind CSS, user focused prototypes, documentation, self service tools
- **Cloud and infrastructure:** Azure, Docker, Kubernetes, Terraform, Airflow, Databricks, Git
- **Educational and media work:** Teaching support, tutorial delivery, slide structure, technical explanation, podcast production, interview preparation, audio editing, research communication

RELEVANT EXPERIENCE

Data & AI Platform Engineer

Robodata

Amsterdam, Netherlands

September 2025 - June 2026

- Built privacy aware AI and data platform components for enterprise clients, including metadata driven ingestion, schema inference, and PII classification workflows.
- Developed automated data pipelines in Python, PySpark, and SQL using Bronze, Silver, and Gold architecture.
- Built local LLM workflows for data discovery, field classification, and metadata generation, with attention to safe and governed use of internal data.
- Created AI ready data interfaces that allowed internal datasets to be queried more safely by LLM based agents.
- Worked on self service data tooling so non specialist users could understand, operate, and reuse technical data infrastructure.
- Documented platform components, workflows, and design choices to support handover, adoption, and future development.
- Used Docker, Kubernetes, Terraform, Azure, Airflow, Databricks, and SQLAlchemy across production platform work.

Teaching Assistant - Student Assistant - Student Ambassador

Vrije Universiteit Amsterdam

Amsterdam, Netherlands

June 2022 - November 2024

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts in a clear and practical way, with focus on reasoning rather than memorising procedures.
- Helped students understand the assumptions behind probability exercises and build confidence with formal problem solving.
- Built a student portfolio system with practical tools and info for first year students and represented the EOR program during Open Days.

Operational Trader

QuantFi

Amsterdam, Netherlands

October 2022 - May 2023

- Developed and tested Python based forecasting workflows using machine learning models, including Random Forest, XGBoost, and neural networks.
- Built data ingestion, retraining, and daily forecast workflows in Python and SQL to support recurring operational decisions.
- Created dashboards and visualisations to compare model output, forecast quality, and realised performance.

EDUCATION

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making • Vrije Universiteit Amsterdam

• Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam

• Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •
Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds • Leeds, England
• 2024 • 8.5/10

PROJECTS

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
 - Next.js, TypeScript, FastAPI, Python, Tailwind CSSBuilt a full-stack portfolio with a FastAPI backend that parses an Obsidian vault into structured API endpoints for notes, quotes and chat functionality. Designed it as a self-service knowledge interface, combining content modelling, backend automation, frontend usability and LLM-based interaction.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
 - Python, FastAPI, ChromaDB, sentence transformers, LLM APIsBuilt a retrieval-augmented chatbot using semantic chunking, vector search, structured prompts and citation-based responses. Focused on reliable retrieval, traceable answers, hallucination reduction and reusable workflow design. Designed the system around the core problem of making a knowledge base queryable by an LLM while preserving source grounding.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
 - HTML, CSS, JavaScriptBuilt and maintain a public website focused on clear navigation, practical usability and simple content management for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics, and philosophy. Prepared technical topics, structured long form conversations, and edited material for clarity and pacing.

RELEVANT STRENGTHS

- Building practical AI tools rather than only experimenting with models.
- Translating unclear workflows into usable technical prototypes.
- Explaining technical and mathematical concepts to students and non specialist users.
- Working across backend, data, AI, automation, and user interface layers.
- Thinking critically about privacy, data access, citations, and responsible GenAI use.
- Improving educational material by making structure, pacing, and explanation clearer.